

Beregningsteknik for Computer Ingeniører

MM 3: Induction and recurrence

Topics:

- Principle of mathematical induction
- Examples of using induction
- Mistaken induction proofs
- Strong induction
- Recursive definitions and recursively defined functions.

Motivation

Some statements are valid for all positive n



n can take any value from
a pool of infinitely many values



How to prove that the statement is valid for all n ?

Principle of mathematical induction

PRINCIPLE OF MATHEMATICAL INDUCTION To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: We verify that $P(1)$ is true.

INDUCTIVE STEP: We show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

Mathematical argument consisting of:

- **A proposition $P(n)$:** A particular statement/function that is asserted true for $k \geq b$ (or for all non-negative or positive integers)
- **A base case/basis step:** Asserting that $P(b)$ is true for the initial value b
- **An inductive hypothesis:** Assume we know $P(k)$ is true for *some* $k \geq b$
- **An inductive step:** "If $P(k)$ is true, we show that $P(k+1)$ is true
($P(k) \rightarrow P(k+1)$): if "P(k) then P(k+1)", or "P(k) only if P(k+1)")
- **Conclusion:** $P(n)$ is true for all positive integers $n \geq b$

Why mathematical induction is valid?

- "Informal" explanation:
 - $P(1)$ is true. Then we know that $P(2)$ is true, because $P(1)$ implies $P(2)$. Further, we know that $P(3)$ is true, because $P(2)$ implies $P(3)$. Continuing along these lines, we see that $P(n)$ is true for every positive integer n .
- Formal explanation:
 - The validity of the principle of mathematical induction follows from a fundamental axiom on the set of integers - the *well-ordering property*:

THE WELL-ORDERING PROPERTY:

Every nonempty set of nonnegative integers has a least element.

Note that...

- We do not necessarily start with $n = 1$, but, if $P(1)$ and $P(n) \rightarrow P(n+1)$ are true for $n \geq 1$, then also $P(b)$ and $P(k) \rightarrow P(k+1)$ for $k \geq b$
- Induction can fail
 - Doesn't mean the property is false (or true)
OBS! Truth value of $P(k) \rightarrow P(k+1)$
- – Choosing what to prove is usually the hardest part
- Mathematical induction does not provide the answer to the problem. However, when we have guessed the answer, we can rigorously prove it – i.e. not a recipe for discovering formulas and relations

Examples

- Example 1. Show that if n is a positive integer, then

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

- Example 2. Show that this statement is true for all nonnegative integers

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$$

- Example 3. Sum of geometric progression
- Example 4. Use mathematical induction to prove that for n greater or equal to 4

$$2^n < n!$$

Examples

- Example 5. An inequality for harmonic numbers
- Example 6. Prove that $n^3 - n$ is divisible by 3 when n is a positive integer
- Example 7: Prove that a set with n elements has 2^n subsets (viz. the previous lecture)

Example

- Example. Derive a formula for the sum of the first n positive odd integers:
 - Testing: $1 = 1$, $1 + 3 = 4$, $1 + 3 + 5 = 9$,
- Conjecture $P(n)$: ?
- Proof by induction that $P(n)$ is true for all positive n :
 - $P(1)$ holds, because?
 - Let's assume $P(k)$ holds, then $P(k+1)$ holds, because ...?
 - What are we to prove?
 - Thus, by mathematical induction, since ...? $P(n)$...?

Rain example

- Consider a planet X, where the following rule holds (is true!):
 “If it rains one day, it also rains the next day”
- You land on planet X and it **does not rain** on the day you arrive. What can you conclude?
 - a) *It will never rain on planet X*
 - b) *It has never rained on planet X*
 - c) *It did not rain yesterday on planet X.*
 - d) *It will rain tomorrow on planet X.*
- You land on planet X and it **does rain** on the day you arrive. What can you conclude?
 - a) *It rains every day on planet X.*
 - b) *It will rain tomorrow on planet X.*
 - c) *It rained yesterday on planet X.*
 - d) *It will rain every day from now on.*

Silly (incomplete) proof

- $P(n)$: $n = n + 1$, n is a positive integer
 - Inductive step – what do we need to show?
 - How should we start and proceed?
- $P(n)$: Every set of n lines ($n \geq 2$) in the plane, no two of which are parallel, meet in a common point